Words to Embeddings Note-Taking Framework

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1 Summary

This week we are talking about how to take words and turn them into embeddings. The main problem trying to be solved is that machines can't interpret plain words, we have to turn those words into a number or more accurately, a vector of numbers in order for the computer to understand/interpret them. There are many methods for doing this and every year there are more strategies being explored and discovered as shown by the tables in the survey: Word Embeddings a Survey. These methods fit into two main groups, prediction based models and count based models.

Prediction based models are derived from neural network language models and they leverage language models to predict the next word (Almeida and Xexeo, 2). Examples of prediction based models include skip-gram and continuous bag-of-words. Both of these models predict words in different ways. Skip-gram tries to predict words around a central word, meanwhile continuous bag-of-words takes the words around a central word and tries to predict the central word (Almeida and Xexeo, 6). Count based models also attempt to predict the next word given context, but the differ from prediction based models because they use word-context matrices instead of language models. These matrices are formed by looking at all of the contexts in which a certain word occurs and then calculating the co-occurrence count between that word and its context words (Almeida and Xexeo, 6). The biggest difference in these models is that prediction based focus on local contexts, while count based models use the whole corpus at once.

2 Attention Grabbers

1. One thing that I really liked in the video was how he had a diagram of the neural network and its activation functions. The visual really helped show the process.

2. The graphing of the words in the video was interesting. It makes sense, similar words should have similar vectors and therefore occupy the same

area of a graph. When you start having a bunch of weights the graph space would be more and more complex.

3. Looking at table 1 and 2 in the Word Embeddings Survey, it's crazy how much changes and improves in a relatively short time. There was a new paper/strategy every year or 2.

4. I thought Sub-Sampling was an interesting concept. It makes sense to avoid common words because they are more easily associated with a large variety of words. (Distributed Representations of Words and Phrases and their Compositionality, 4)

5. 33 billion words is crazy, I didn't even know there were that many words! It did get 72% accuracy with this data compared to 66% with 6 billion words. This is a lot better than the paper we looked at last week which got around 40% accuracy. (Distributed Representations of Words and Phrases and their Compositionality, 7)

3 Questions

1. In the video, we start with random weights and then we get new weights that work really well. Where did the new weights come from?

2. These talk about softmax a lot, what exactly is softmax and how does it work?

3. What does corpora mean?

4. How does backpropagation work?

5. Are there certain situations where count based models are better than prediction based models and vice versa?